

JOSE ANTONIO ELIZALDE

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Education

University of California, Los Angeles

Sept 2025 – June 2027

B.S. in Computer Science and Engineering

Westwood, CA

Los Angeles Pierce College

Feb 2023 – June 2025

A.A. in Physics and Mathematics, GPA: 3.92/4.00

Woodland Hills, CA

Relevant Coursework

Data Structures and Algorithms, Networking Fundamentals, Distributed Systems, Operating Systems Architecture, Systems Programming, Computer Architecture, Discrete Mathematics, Object-Oriented C++

Technical Skills

Languages: C++, C#, C, Python, Bash

Tools & Frameworks: Linux, FastAPI, Pandas, Unity, Vim/NeoVim, CMake, Git/GitHub, LaTeX

Experience

University of California, Los Angeles – SURE C² Research Program

June 2024 – August 2024

Undergraduate Researcher, Department of Mathematics

Los Angeles, CA

- Conducted research in combinatorics on partially ordered sets (posets), studying transfer systems and their patterns.
- Developed a C++ program to enumerate all transfer systems of a given finite poset, automating computations for the amount of transfer systems for a given set, where it can grow exponentially with the cardinality of the set.
- Discovered a recursive relationship for the family of sets X_n^{++} , contributing to a co-authored research paper presented at the University of Texas at Arlington.

Carnegie Mellon University – REUSE Research Program

May 2025 – Aug 2025

Undergraduate Researcher, Human-Computer Interaction Institute (HCII)

Pittsburgh, PA

- Developed an interactive application to investigate methods for improving communication and empathy between rideshare drivers and passengers.
- Implemented and iteratively refined application features based on user feedback and usability testing conducted throughout the research process.
- Collaborated with a faculty advisor, graduate students, and a research partner to design and improve the system.
- Co-authored a research paper accepted to CHI 2026 evaluating the application's impact on user engagement and empathy.

Projects

RobinHood Discord Portfolio Bot | C++, Python, FastAPI, Pandas, D++, Linux

June 2026 – Present

- Developed a C++ Discord bot using the D++ library that interfaces with a Python FastAPI backend to provide real-time Robinhood portfolio analytics and stock screening.
- Implemented a background market data pipeline using `yfinance` and Pandas to periodically download and process historical S&P 500 data, computing momentum breakouts and valuation metrics including P/E ratio and ROE.
- Deployed the application on a DigitalOcean Linux VPS and configured a `systemd` service for automatic startup, crash recovery, and 24/7 availability.

Jump the Gun – UCLA ACM Studio | C#, Unity, Git

October 2024

- Led the design and development of a first-person action shooter, defining gameplay mechanics and coordinating implementation across the team.
- Implemented a custom physics-based movement controller in C#, replacing Unity's default character controller with manually programmed movement and collision logic.
- Collaborated through Git branching and pull requests to integrate gameplay features developed concurrently by multiple team members.

KurtCobot | C++, Arduino, Embedded Systems

Sep 2024 – Oct 2024

- Developed an embedded C++ application using an Arduino that performs programmed musical notes.
- Implemented pulse width modulation (PWM) signal generation to convert digital note sequences into precisely timed audio output.
- Designed and assembled the supporting electronic circuitry, integrating embedded software with custom hardware using breadboards and soldered components.